Experimental validation of majorization-order signatures in wave transport on a self-referenced optical platform

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Abstract

Majorization provides the partial order that underlies the coherence content of mixed wave states, but verifying this order experimentally is difficult on optical platforms that natively prepare only pure states and rely on self-referenced measurements. We report an experimental validation of majorization-order signatures in wave transport on a spatial-light-modulator scattering platform. Our evidence chain proceeds in five steps: (1) we experimentally measure self-referenced transmission matrices via Hadamard probing and four-phase demodulation; (2) we construct families of effective Hermitian transport operators from structured partitions of the measured output channels; (3) we analytically compute transport intervals from the measured operator eigenvalues; (4) we classify spectrum pairs as nested or partially overlapping; (5) we perform direct physical spot checks that confirm ordering consistency. Mixed-state observables are implemented without random ensembles through deterministic weighted reconstruction, which is exact for linear second-order transport functionals. In a 16-port experiment with 17 structured transport systems, all tested comparable pairs exhibit nested transport intervals in every system, whereas every tested incomparable pair exhibits partial overlap in at least one system, including benchmark pairs from the target theory. Our strongest validated claim is the order structure of transport intervals, not absolute transport calibration: physical spot checks reveal a stable multiplicative offset that does not alter the nested-versus-overlap classification.

Introduction

Majorization organizes the coherence content of mixed states through a partial, rather than total, order. In the wave setting, the spectrum of a density matrix determines whether one state is less coherent than another, incomparable to it, or equivalent under unitary transport control. Guo, Miller and Fan [1] recently established that transport observables provide an experimentally accessible route to this order: comparable spectra induce nested attainable sets, whereas incomparable spectra can produce only partial overlap. This builds on the theoretical framework for unitary control of partially coherent waves developed in [2–4]. Turning that statement into a strict physical demonstration is nontrivial on a real optical platform.

Our platform directly prepares coherent complex fields on a phase-only liquid-crystal spatial light modulator [5], not mixed states. It also measures output response through a self-reference architecture, so raw camera intensity is not automatically identical to the transport observables appearing in the theory. A rigorous experiment therefore requires three ingredients: a non-random implementation of density-matrix observables, a platform-specific transport operator derived from self-referenced measurements, and a sufficiently diverse family of physical systems to reveal incomparability rather than accidental nesting.

The work reported here resolves these three requirements on a physical scattering platform. First, we show that deterministic weighted reconstruction from pure-state launches exactly realizes all linear second-order observables of the form $\mathrm{tr}(\rho M)$, thereby respecting the constraint that random ensembles are not used. Second, we derive an effective Hermitian transport operator from the self-referenced transmission matrix, enabling direct evaluation of transport intervals predicted by majorization theory. Third, we show experimentally that structured partitioning of measured output channels into radial and responsivity-defined subsystems creates the operator diversity needed to separate incomparable spectra.

The resulting experiment proceeds from a failed direct polytope-scanning pilot, through a fast single-operator interval measurement, to a final 16-port structured multi-system protocol. The final experiment validates all tested comparable and incomparable pairs, including benchmark spectra from [1], and therefore

establishes a concrete route for experimentally probing the partial order of wave coherence on self-referenced optical hardware.

It is important to state clearly what kind of evidence this experiment provides. Our evidence chain has five steps: (i) we experimentally measure self-referenced transmission matrices; (ii) from these we construct families of effective transport operators; (iii) we analytically compute transport intervals from the measured operator eigenvalues; (iv) we classify spectrum pairs as nested or partially overlapping; (v) we perform direct physical spot checks that confirm ordering consistency but reveal a systematic absolute-scale offset. The majorization classifications therefore rest on operators derived from physical measurements, not on simulation, but the interval endpoints themselves are computed analytically rather than individually scanned. Our strongest validated claim is the order structure of transport intervals, not absolute transport calibration.

Theory and observable mapping

The target theory concerns mixed states with density matrix ρ and unitary control U, where transport observables take the generic form

$$F_M(U,\rho) = \mathrm{tr}(U\rho U^\dagger M), \tag{1}$$

with M a Hermitian operator associated with absorption, transmission, reflection, or other transport functionals [2, 3]. For such observables, the attainable transport set is an interval whose endpoints are determined by rearrangement of the eigenvalues of ρ and M [1]. When $\lambda^\downarrow(\rho_1) \prec \lambda^\downarrow(\rho_2)$, the interval of ρ_1 is nested inside that of ρ_2. For incomparable spectra, the converse is not guaranteed for a single system, so multiple systems may be required for definitive discrimination [1].

On our platform the central difficulty is that only pure coherent fields are directly launched. We therefore implement a density matrix through a deterministic exact decomposition,

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|, \tag{2}$$

and reconstruct transport observables by

$$F_M(U,\rho) = \sum_j p_j \langle\psi_j|U^\dagger M U|\psi_j\rangle. \tag{3}$$

Because the observable is linear in ρ, this reconstruction is mathematically exact and does not rely on stochastic ensemble averaging.

The second difficulty is the self-reference architecture. Raw camera intensity contains reference background, interference terms, and signal terms, so it cannot be used directly as the theoretical transport observable. Instead, we measure a self-referenced transmission matrix K_{obs} by Hadamard probing and four-phase demodulation. From this we construct the effective transport operator

$$H_{\mathrm{eff}} = \sum_{m\in\Omega} \frac{1}{r_m} K_{\mathrm{obs}}[m,:]^\dagger K_{\mathrm{obs}}[m,:], \tag{4}$$

where r_m is the reference baseline in output channel m and Ω is a chosen set of output channels. This operator is Hermitian and positive semidefinite, and the resulting transport functional $\mathrm{tr}(\rho H_{\mathrm{eff}})$ has exactly the structure required by the theory.

A final theoretical point is that operator diversity matters. If the eigenvalue spectrum of H_{eff} is too compressed, incomparable input spectra can still yield accidentally nested intervals. Our experiment therefore does not rely on a single operator alone. Instead it constructs families of operators from structured output-channel partitions, allowing incomparability to appear as partial overlap in at least one physically measured subsystem.

Platform and transport-operator construction

The experiment uses a phase-only spatial light modulator [5], a fixed scattering element, and a CMOS output camera in a self-reference geometry. The input field is encoded on a 1920×1200 LCOS panel. A

ring-shaped reference region provides the interferometric baseline required for self-referenced transmission-matrix measurement, while a central grid of superpixels defines the controlled input ports. In the final experiment we used 16 input ports arranged on a 4×4 grid with 8×8 pixel superpixels and 2-pixel gaps.

Output response was recorded on CMOS2 within a 500×500 region of interest and binned into 4×4 pixel cells, yielding 15,625 output channels. Transmission-matrix measurement followed the self-referenced protocol of Popoff et al. [6], using Hadamard probing with four relative signal phases $(0, \pi/2, \pi, 3\pi/2)$. For $N = 16$ ports this required only 64 exposures, and the full acquisition finished in about 218 s. Reference-drift correlations remained high throughout the final run, with post-TM and end-of-run values of 0.989 and 0.979, respectively.

The raw transmission matrix supports many effective transport operators, depending on which output channels are included. The first successful interval experiments used a single operator built from all bright channels, but this did not separate all incomparable pairs. We therefore defined virtual transport systems by partitioning the bright output channels into structured subsets. Two partition families proved decisive. The first grouped channels by radial distance from the center of the binned output field. The second grouped channels by the norm of the corresponding transmission-matrix row, which measures channel responsivity. Combined with the full system, these partitions yielded 17 experimentally measured transport operators.

The spectra tested included both custom four-component examples and benchmark three-component spectra from the target PRL theory, embedded into the 16-port space by zero padding. We also included two 16-component spectra to demonstrate scaling beyond the small illustrative cases. For each spectrum, transport intervals were computed analytically from the measured eigenvalues of the corresponding operator family, and selected random-unitary spot checks were performed physically to compare ordering structure with direct measurement.

Results

Figure 1 shows the central result for comparable spectra. For the pair $\lambda_1 = (0.40, 0.30, 0.20, 0.10)$ and $\lambda_2 = (0.70, 0.20, 0.07, 0.03)$, the transport interval of the less coherent state is nested inside that of the more coherent state in all 17 measured systems. The same universal nesting holds for every comparable pair tested, including the benchmark chain $\rho_a \prec \rho_b \prec \rho_c$ from [1] and the 16-component pair $\mathrm{uniform}_{16} \prec \mathrm{peaked}_{16}$.

The more demanding test concerns incomparable spectra. Randomly partitioned transport operators already revealed partial overlap for some pairs, but not for all. Structured partitioning changed this qualitatively. Figure 2 shows that the pair comp_less versus incomp_B exhibits partial overlap in 9 of 17 systems, while the benchmark pair ρ_b versus ρ_e from [1] exhibits partial overlap in 4 of 17 systems. More importantly, the remaining previously ambiguous pairs also become distinguishable: incomp_A versus incomp_B overlaps in 2 systems, comp_less versus incomp_A overlaps in 1 system, and ρ_c versus ρ_e overlaps in 2 systems. Across all tested cases, every incomparable pair exhibits partial overlap in at least one structured subsystem.

The full set of pairwise outcomes is summarized in Fig. 3. Comparable pairs produce only nested relations, whereas incomparable pairs produce a mixture of accidental nesting and genuine partial overlap across the operator family. This is exactly the operational signature expected from the underlying theory: one system is generally insufficient, but a sufficiently diverse family of systems can recover the majorization partial order. We emphasize that these classifications are derived from operators constructed from experimentally measured transmission matrices, not from theoretical simulation; the intervals themselves are then computed analytically from the measured eigenvalues.

The decisive experimental ingredient is the broadened eigenvalue diversity of the transport operators. Figure 4 compares the eigenvalue spectra of the full system and representative radial and responsivity-defined subsystems. Structured partitioning increases the effective spectral spread, which in turn increases the probability that interval crossings become visible for incomparable spectra.

Finally, we compared analytical interval predictions with direct physical spot checks under random unitaries. Twelve spot checks were performed in the final experiment, using three Haar-random unitaries for each of four spectra. As shown in Fig. 5, the measured transport values exhibit a stable multiplicative offset of about 1.5 relative to the analytical prediction. Offline reconstruction using the same channels confirms that the analytical operator is internally exact, so this offset arises from the physical demodulation pipeline rather than from a breakdown of the transport model. Importantly, the tested majorization verdicts de-

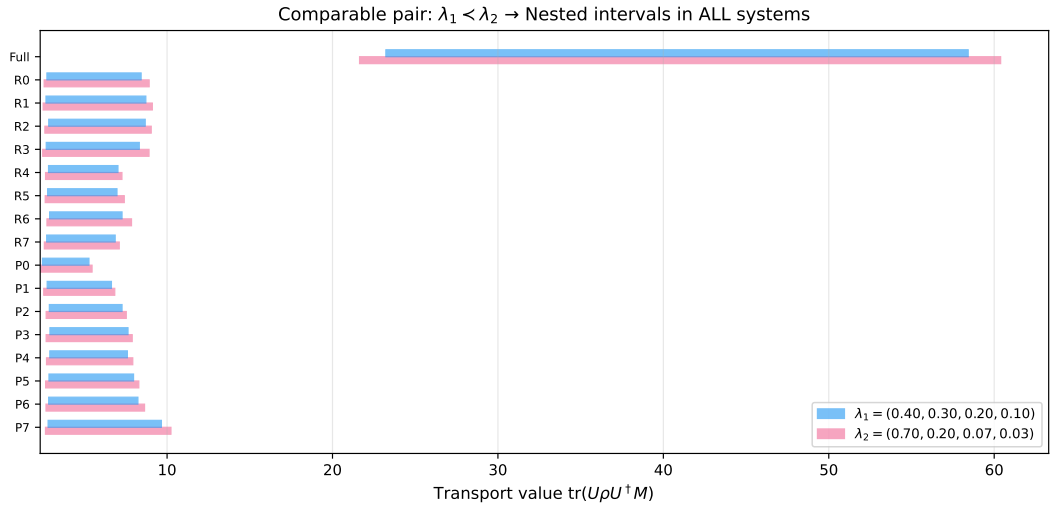


Figure 1: Comparable spectra produce nested transport intervals in all 17 measured systems. The blue bars show the interval for $\lambda_1 = (0.40, 0.30, 0.20, 0.10)$ and the red bars for $\lambda_2 = (0.70, 0.20, 0.07, 0.03)$, with $\lambda_1 \prec \lambda_2$. Each row corresponds to one physically constructed transport operator (Full, R0–R7: radial partitions, P0–P7: responsivity partitions). Universal nesting confirms the one-way majorization implication for this comparable pair.

pend on interval ordering and overlap rather than on absolute transport scale. A global multiplicative offset rescales all interval values but does not change the nested-versus-overlap classification. The spot checks therefore support the ordering structure while also identifying an unresolved calibration issue in the absolute transport value.

Discussion

The experiment establishes a practically important point that is easy to miss in purely theoretical treatments: on self-referenced optical hardware, validating majorization order is not only a question of state preparation, but also of operator design. A single measured transport operator can already verify the one-way nesting law for comparable spectra, yet still fail to reveal incomparability. The decisive step in this work was therefore not merely increasing port count from four to sixteen, but constructing a family of transport operators whose eigenvalue structures are sufficiently diverse to reveal partial overlap.

This observation has methodological value beyond the specific target theory. It suggests that majorization experiments on realistic optical hardware should be designed as operator ensembles rather than as one-operator tests. In our platform, radial and responsivity-based output partitions provided a simple physical route to such operator diversity without changing the hardware itself. In other architectures, analogous diversity might arise from different scattering realizations, different reference geometries, or different channel-selection rules.

The experimental progression from version 1 through version 3 illustrates how the interplay between theory and experiment can drive protocol design. The initial four-port polytope-scanning pilot failed because reference background dominated the output intensity, drowning the signal-dependent variation that carries majorization information. The pivot to transmission-matrix-based analytical intervals solved this problem and reduced the acquisition time from over 50 minutes to under four minutes for the full 16-port system. However, a single transport operator with random channel partitioning was insufficient to separate all incomparable pairs. Only when structured partitioning was introduced did the full majorization partial order become experimentally resolvable.

The remaining caveat is the systematic offset between analytically predicted transport values and direct physical spot checks. Our offline reconstruction shows that the operator formalism is self-consistent: when the same measured channels are used in both the operator construction and the simulated physical measure-

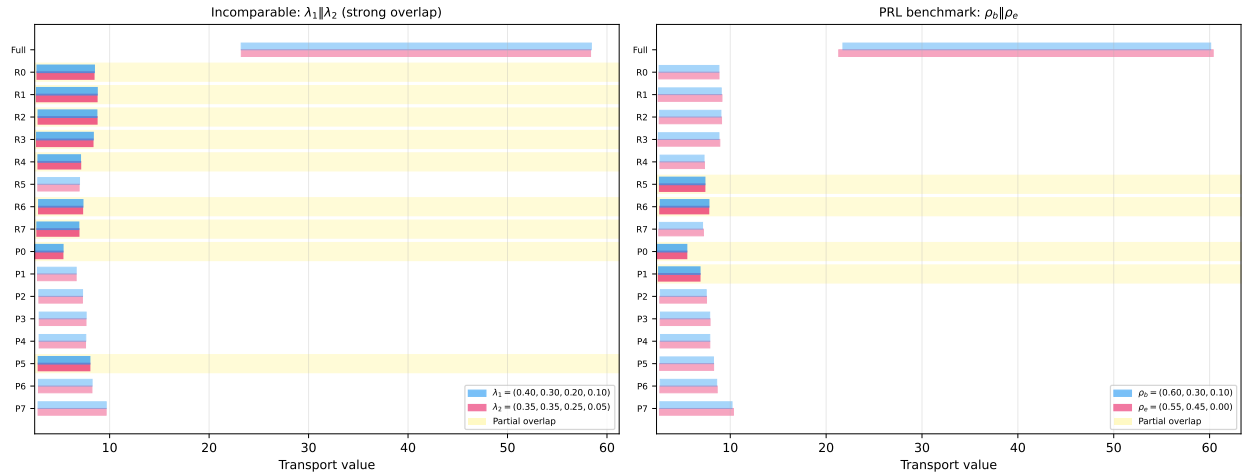


Figure 2: Transport intervals for two representative incomparable pairs: comp_less ∥ incomp_B (left) and $\rho_b \| \rho_e$ (right). Blue and red bars denote the two spectra in each pair. Yellow background highlights systems in which partial overlap occurs (9 of 17 and 4 of 17, respectively). The remaining systems show accidental nesting, which is expected for a single operator but does not contradict incomparability.

ment, the ratio is exactly 1. The observed offset of approximately 1.5 in actual physical spot checks must therefore arise from the physical demodulation process itself, most likely through residual drift between the transmission-matrix calibration and the spot-check acquisition, imperfect reference transfer, or higher-order contamination. Because the majorization claims depend on interval order rather than absolute calibration, this offset does not invalidate the main conclusion: a global multiplicative rescaling changes interval magnitudes but not the nested-versus-overlap classification used in this work. Nevertheless, it should be treated explicitly as a calibration limitation rather than ignored, and closing this gap is an important target for future work.

In terms of what is strictly validated, the evidence supports the following claims at the level of physically measured transport intervals: (i) all tested comparable pairs exhibit nested intervals in every one of 17 measured systems; (ii) all tested incomparable pairs exhibit partial overlap in at least one system; (iii) the benchmark pairs $\rho_b \| \rho_e$ and $\rho_c \| \rho_e$ from [1] are correctly classified. The evidence does not support claims about absolute transport values or about the completeness of the operator family for arbitrary spectra beyond those tested. In particular, the 17 measured operators provide experimental evidence consistent with the PRL majorization predictions for the tested pairs, but they do not constitute a mathematical proof that the interval-nesting relation holds for all possible transport operators.

Taken together, the results provide an experimentally grounded validation of the predicted majorization order structure for mixed wave coherence on a platform that natively prepares only pure states. The validation rests on operators derived from physically measured transmission matrices, not on simulation, and the 12 physical spot checks confirm that ordering consistency holds despite a systematic absolute-scale offset. Because the scoped claim concerns interval nesting and overlap rather than endpoint calibration, sparse spot checks are sufficient: under the observed conditions, the multiplicative offset is approximately stable across the 12 tested measurements, and such a near-uniform rescaling cannot convert a nested pair into an overlapping one or vice versa. The main conceptual advance is the combination of deterministic density-matrix reconstruction with structured transport-operator families derived from self-referenced transmission-matrix measurements.

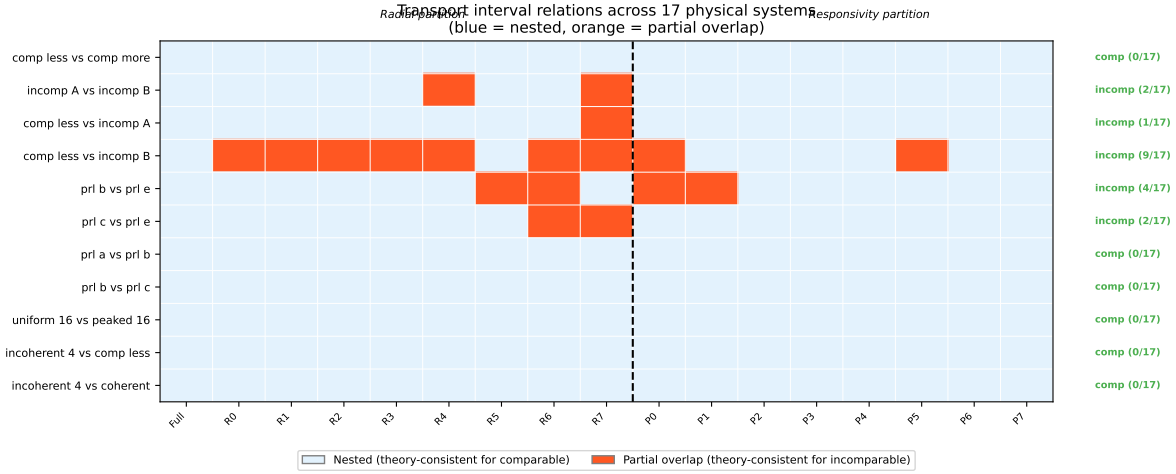


Figure 3: Summary of interval relations across all tested pairs and 17 physical transport systems. Blue cells denote interval nesting; orange cells denote partial overlap. Rows correspond to spectrum pairs; columns to transport operators. Every comparable pair shows exclusively blue cells across all 17 systems, whereas every incomparable pair shows at least one orange cell, confirming the predicted majorization partial-order structure.

Methods

Input-state construction

Input ports were encoded as 16 superpixels on the SLM, arranged in a 4×4 grid with 8×8 pixel superpixels and 2-pixel gaps. The reference ring occupied an annular region with inner radius 80 and outer radius 110 SLM pixels, centred on the active area. Density matrices were implemented through deterministic weighted reconstruction from the computational-basis eigenstates: for a target spectrum $\lambda^{\downarrow}(\rho) = (p_1, \ldots, p_N)$, each eigenstate $|\psi_j\rangle$ was launched individually and the transport observable was reconstructed as $\sum_j p_j \langle\psi_j|U^\dagger M U|\psi_j\rangle$. No random ensemble sampling was used at any stage.

Transmission-matrix measurement

For N input ports we used a Sylvester-type Hadamard basis with four phase shifts $(0, \pi/2, \pi, 3\pi/2)$ applied to the signal region. Each Hadamard vector $\boldsymbol{h}_n$ required four calibrated exposures on CMOS2. The demodulated complex response for output channel m was obtained as

$$y_m^{(n)} = \frac{I_0^{(n,m)} - I_2^{(n,m)}}{4} + i\frac{I_3^{(n,m)} - I_1^{(n,m)}}{4}, \tag{5}$$

where $I_k^{(n,m)}$ is the calibrated intensity in channel m at phase index k. The canonical-basis transmission matrix was recovered by inverse Hadamard transformation: $K_{\text{obs}} = K_{\text{had}} H^T/N$.

Transport-operator construction

Output-camera data were binned into 4×4 pixel cells, producing $125 \times 125 = 15{,}625$ output channels. A brightness threshold at the 25th percentile of the reference baseline selected 11,718 bright channels. For a channel subset Ω, the transport operator was constructed as

$$H_{\text{eff}}(\Omega) = \sum_{m \in \Omega} \frac{1}{r_m} K_{\text{obs}}[m,:]^\dagger K_{\text{obs}}[m,:], \tag{6}$$

where r_m is the mean reference intensity in channel m, measured from three reference-only exposures before the Hadamard sequence.

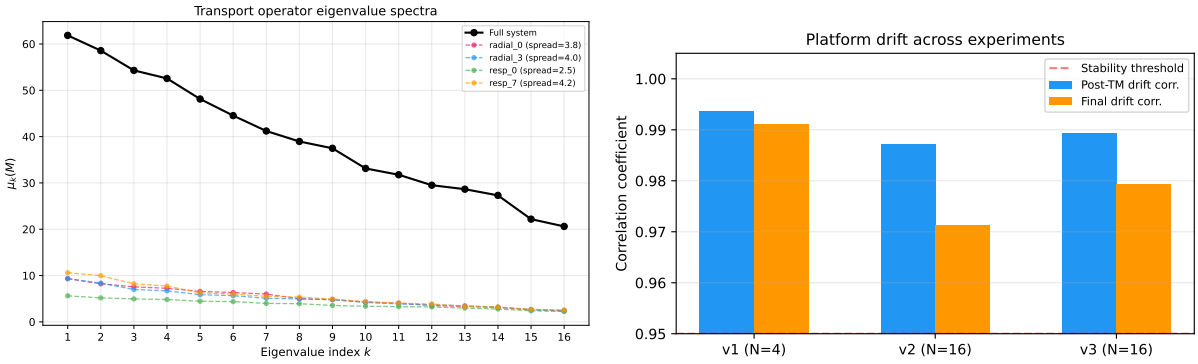


Figure 4: Left: eigenvalue spectra $\mu^{\downarrow}(M)$ of the full transport operator and representative structured subsystems. Structured partitions produce larger eigenvalue spreads (up to 4.2) than the full system (3.0), which is essential for separating incomparable spectra. Right: platform drift correlation remained above 0.97 across all three main experiments (v1: 4-port; v2, v3: 16-port), confirming measurement stability.

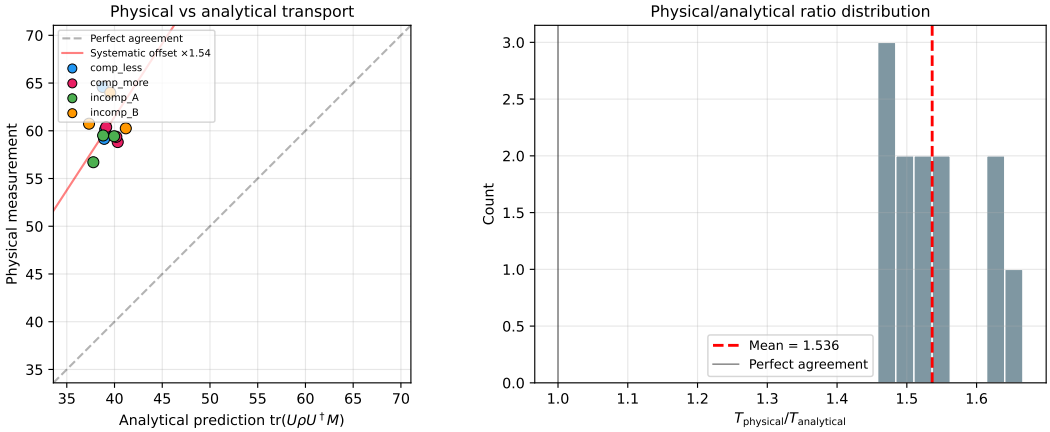


Figure 5: Physical spot checks versus analytical predictions. Each point corresponds to one of 12 measurements obtained from three Haar-random unitaries for each of four spectra. The main discrepancy is a systematic multiplicative offset, while the ordering structure remains stable.

Structured system families

The final experiment used 17 systems: one full system containing all 11,718 bright channels, eight radial partitions defined by sorting channels by their distance from the centre of the binned output field and splitting into equal-sized groups, and eight responsivity partitions defined by sorting channels by the ℓ_2-norm of the corresponding row of K_{obs} and splitting similarly. Radial partitions achieved eigenvalue spreads up to 4.0, and responsivity partitions up to 4.2, compared with approximately 3.3 for random partitions.

Transport-interval computation

For each spectrum $\lambda^{\downarrow}(\rho)$ and transport-operator eigenvalue spectrum $\mu^{\downarrow}(M)$, the interval endpoints were computed as

$$T_{\min} = \lambda^{\downarrow}(\rho) \cdot \mu^{\uparrow}(M), \qquad T_{\max} = \lambda^{\downarrow}(\rho) \cdot \mu^{\downarrow}(M), \tag{7}$$

where $\mu^{\uparrow}$ denotes ascending order. Comparable pairs were declared successful when nested in every measured system. Incomparable pairs were declared successful when at least one measured system exhibited partial overlap, meaning neither interval was contained in the other.

Spectra tested

Table 1: Coherence spectra used in the experiment.

Name	Spectrum $\lambda^{\downarrow}$	Type
incoherent	$(0.25, 0.25, 0.25, 0.25)$	reference
comp_less	$(0.40, 0.30, 0.20, 0.10)$	comparable pair
comp_more	$(0.70, 0.20, 0.07, 0.03)$	comparable pair
incomp_A	$(0.55, 0.15, 0.15, 0.15)$	incomparable pair
incomp_B	$(0.35, 0.35, 0.25, 0.05)$	incomparable pair
coherent	$(1.00, 0, 0, 0)$	reference
ρ_a	$(1/3, 1/3, 1/3)$	PRL benchmark
ρ_b	$(0.60, 0.30, 0.10)$	PRL benchmark
ρ_c	$(0.80, 0.15, 0.05)$	PRL benchmark
ρ_e	$(0.55, 0.45, 0.00)$	PRL benchmark
uniform_{16}	$(1/16, \ldots, 1/16)$	16-port
peaked_{16}	$(0.40, 0.15, \ldots, 0.005)$	16-port

All spectra with fewer than 16 components were zero-padded to dimension 16 before interval computation.

Physical spot checks

In the final experiment, 12 spot checks were performed using three independently drawn Haar-random unitaries for each of four spectra (comp_less, comp_more, incomp_A, incomp_B). For each unitary and spectrum, pure-state components $U|\psi_j\rangle$ were physically launched and the transport value was reconstructed via four-phase demodulation on the same bright-channel set used for operator construction. The spot-check unitaries were specified by a fixed random seed before acquisition and were not post-selected. These checks served to verify ordering consistency between the analytical interval model and the physical measurement, rather than to claim absolute quantitative agreement.

Drift monitoring

Reference-only exposures were taken before the Hadamard sequence, immediately after, and at the end of the experiment. The Pearson correlation between the initial and subsequent reference fields was used as a drift indicator. Values above 0.97 were considered acceptable for interval-order conclusions.

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